Artificial Intelligence

BSc Computer Science - Durham University

Matthew Emerson

November 2026

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1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	9th Oct	15th Dec	21st Jan	Dr Yang Long	50%
AI Search	15th Jan	18th Feb	18th Mar	Prof Iain Stewart	25%
Bias in AI	5th Feb	30th Mar	6th May	Dr Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	-
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, M.P., Faisal, A.A., and Ong, C.S., 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, G., 2017).

2.1 Introduction to Machine Learning

2.1.1 Module Information

This module is organised around a direct connection between teaching and practical work. The lectures introduce the concepts, mathematics, assumptions, and experimental consequences needed for that week's practical. The practical then produces the evidence used in the corresponding coursework report. This can be seen in Table 3.

Stage	What you do	What you keep
Lecture	Understand the main question, mechanisms, assumptions, and key equations.	Notes in your own words and questions that still need an answer.
Practical	Implement, test, compare, visualise, and discuss the method.	A notebook that runs from top to bottom with outputs included.
Weekly Report	Select the strongest evidence and explain why the result occured.	A concise report that links theory, method, result, and interpretation.
Portfolio	Revise the nine reports and connect ideas across the module.	A coherent record of technical progress and developing judgement.

Table 3: The learning and coursework model.

For more information about expectations for evidence, see (Long and Durham University, 2026, pp.3-4).

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2.1.2 What is Machine Learning?

Definition 2.1.2.1: Arthur Samuel said in 1959 that machine learning is the

> "field of study that gives computers the ability to learn without being explicitly programmed."

The field arose from the question – can a computer learn on its own in order to perform a specific task, rather than relying on a programmer explicitly setting up rules for it? Similarly, Tom Mitchell defined machine learning in 1997 as

> "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

From this, we can define a well-defined learning task as $< T, P, E >$.

There are three typical types of machine learning, these are:

- **Definition 2.1.2.2**: Supervised learning seeks to learn the mapping (rules) between inputs and outputs. Labelled data (correct answers) is provided in past input and output pairs during the learning process to train the model on how it should behave for previously unseen data.
- **Definition 2.1.2.3**: Unsupervised learning seeks to discover the hidden pattern from a set of inputs. Unlabelled data is provided as the past input during the learning process. The program will then attempt to group similar instances into groups.
- **Definition 2.1.2.4**: Reinforcement learning works by providing occasional positive and negative feedback to reinforce behaviours. Good behaviours are rewarded with treats and therefore become more common whereas bad behaviours are punished and therefore become less common. This is a balance between exploration (of unchared territory) and exploitation (of current knowledge).

2.1.3 Differences between AI, ML, and DL

Definition 2.1.3.1: Artificial intelligence is defined as human intelligence being exhibited by machines.

Definition 2.1.3.2: In simple terms, machine learning can be defined as an approach to achieve artificial intelligence.

Definition 2.1.3.3: Deep learning is defined as a technique for achieving machine learning. As such, we can consider AI to be a subset of ML and ML to be a subset of DL.

2.2 Machine Learning Workflow and Coding Environment

2.2.1 Machine Learning Workflow

Definition 2.2.1.1: To train a model, you use data to create a model and fine-tune it to make better predictions. We can then use this predictive model to make predictions on previously unseen data in order to perform specific tasks.

The machine learning workflow is set out as shown below:

1. **Definition 2.2.1.2**: Project Framing: When setting out to create a model, we must first frame the problem. Here, we set out a clear objective that we want the model to achieve. We then define inputs and outputs and establish success metrics (e.g., prediction accuracy).
2. **Definition 2.2.1.3**: Data Preparation: The next step is to prepare the data. To do that, we identify features (inputs) and a label (output). We then carry out data cleaning on the dataset and split the data into training and test sets.
3. **Definition 2.2.1.4**: Algorithm Selection: Next, we select the algorithm we want to use for the model based on the dataset and the objective.

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4. **Definition 2.2.1.5**: Model Training: The training data is used to incrementally improve the model's performance. The progress made in this step is dependant on the quality of the work done in the previous two stages.
5. **Definition 2.2.1.6**: Model Testing: The next step is to take the model and test it against previously unseen data (the test set).
6. **Definition 2.2.1.7**: Hyperparameter Tuning: We can then attempt to improve the training process to further increase the model's accuracy.
7. **Definition 2.2.1.8**: Inference/Prediction: Finally, the model can be deployed for the task outlined in the first step.

2.2.2 Coding Environment

When programming, we traditionally set out the features as variable X and the target label as Y.

2.3 Data Understanding

Definition 2.3.0.1: A sample is the unit for which the model recieves features and, during training, a label.

We then set up a feature matrix X where each row is a different sample and each column is a new feature and y is a matrix where each row is sample's label corresponding to the sample's features in X.

Definition 2.3.0.2: Nearest-centroid classification attempts to classify the samples by setting out one centre for each species and then classifying them based on their nearest centroid. The formula for calculating this is:

$$\mu_c = \left(\frac{1}{N_c}\right) \sum_i : y^{(i)} = cx^{(i)} \tag{1}$$

For each species c, average the feature vectors belonging to that species. Then, subtract the centroid from the new example, square the differences, add them, then take the square root, as shown below.

$$d(x, \mu_c) = ||x = \mu_x||_2 \tag{2}$$

To predict a sample's species, compute the distances from each species' centroid and choose the class with the smallest distance.

$$\hat{y} = arg\ min_c||x - \mu_c||_2 \tag{3}$$

2.4 Data Preparation Workflow

Definition 2.4.0.1: To prepare data for machine learning, we use the data preparation workflow which follows the steps as below:

1. Read and inspect the data.
2. Clean and split the data into train and test sets.
3. Fit the baseline and model.
4. Test and compare the model against the baseline.
5. Scale the model and predict unseen data.

Definition 2.4.0.2: We can standardise our dataset using training-set standardisation to reduce the unbalanced impact features with large average Euclidean distances have on predictions. To do this, we use (4).

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \tag{4}$$

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2.5 Data Construction

2.5.1 Analysing the Raw Data

Definition 2.5.1.1: Data can be considered skewed to the right if the majority of the data lies to the left with a long tail to the right. To measure this, we use a measure called skewness which can be calculated with (5)

$$\text{Skewness} = \frac{3 \times (\text{Mean} - \text{Median})}{\text{Standard Deviation}} = \frac{3(\mu - Mdn)}{\sigma} \tag{5}$$

Definition 2.5.1.2: Datasets may sometimes contain capped features. This can be identified by graphing the data. If there is a spike on the maximum-value, then it is likely that the data is capped.

2.5.2 Sampling

When deciding how large of a sample to take from a population, you have to balance the size as a larger sample reduces random sampling noise, but also increases processing time.

Definition 2.5.2.1: The Kolmogorov-Smirnov test compares the distributions of two samples. It returns the maximum vertical distance between two cumulative distribution functions.

$$D = sup_x |F_{sample}(x) - F_{population}(x)| \tag{6}$$

It can then be used to calculate a p-value measuring the evidence against equal distributions. This means that if it is above 5%, then there is insufficient evidence of a difference in the distributions.

Definition 2.5.2.2: Similarly, for categorical data, you can use the Chi-square goodness of fit test which compares the expected and observed counts of each unique value.

$$\chi^2 = \sum_j \frac{(O_j - E_j)^2}{E_j} \tag{7}$$

where O_j is the observed count and E_j is the expected count.

2.5.3 Building Trustworthy Evaluation Sets

A row should only serve one role. Either as a training set (to fit parameters), a validation set (to compare choices), or as a test set (to get a final accuracy estimate).

Random splitting preserves population proportions only in expectation. A finite test set can contain too many or too few examples from an important region. Rare strata show the largest relative variation.

2.6 Data Transformation

2.6.1 Numerical Transforms

Definition 2.6.1.1: In min-max normalisation, the training minimum becomes 0 and the training maximum becomes 1. Values between them retain their relative position in the observed training range.

$$x' = \frac{x - min_{train}}{max_{train} - min_{train}} \tag{8}$$

Definition 2.6.1.2: In standardisation, values have the training mean subtracted, and then divided by the training standard deviation. The transformed training column has mean near 0 and standard deviation near 1.

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \tag{9}$$

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Definition 2.6.1.3: The estimator interface has three functions: fit (to learn the statistics), transform (to apply statistics), and fit_transform (combines on training). It is also used in imputers, encoders, ColumnTransformer, and Pipeline.

Definition 2.6.1.4: Data leakage occurs when preprocessing learns from rows that should remain unseen. Test-set statistics carry information about the future evaluation distribution back into training.

2.6.2 Skew and Bins

To reduce the impact of skewed columns, they can be log scaled to compress multiplicative differences into smaller additive differences.

$$x' = \log(1 + x) \tag{10}$$

The transformed distribution spreads common values while compressing the tail.

If you use equal-width bins, you may end up with sparse bins if data is skewed. To fix this, we use quartile bins to keep roughly the same number of rows in each bin.

2.6.3 Categories and Missing Values

Definition 2.6.3.1: If a column is unordered, then we can apply a one-hot representation where each unique value in the column becomes a new column with a boolean value of whether it is present. This is done with a one-hot encoder which learns categories during fit and then applies the same column order during transform.

If you have a column with missing data, you can take one of two approaches:

- Delete the incomplete rows - this does remove observations and can impact the data distribution.
- Impute a fitted statistic - preserves row count and creates a value for future rows (normally median).

2.6.4 A Leakage-Free Preprocessing Pipeline

Column-specific preprocessing uses ColumnTransformer to take each column and applies median imputation and standardisation for numeric columns, and categorical encoding for categorical ones.

The end-to-end data flow becomes: taking the raw DataFrame and fit preprocessing, then taking the resulting matrix and training the model.

2.7 Machine Learning Algorithms

2.7.1 What a Learning Algorithm Contains

Definition 2.7.1.1: A learning algorithm consists of the five parts:

- The representation defines possible predictions.
- The parameters.
- The objective scores each parameter choice.
- The optimiser searches.
- The Hyperparameters control the search rather than the fitted relationship.

Definition 2.7.1.2: A representation states which relationships the model can express. A straight line can change its intercept and slope. It cannot bend, create a threshold, or model separate local patterns.

Definition 2.7.1.3: Parameters determine the fitted line. The algorithm estimates them from the training data. Whereas, the hyperparameters such as learning rate α and iteration count control optimisation. The algorithm does not estimate them through the same update rule.

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Definition 2.7.1.4: The objective converts a complete set of predictions into one number. A lower cost means the current parameter values match the training observations more closely under the chosen definition of error.

Definition 2.7.1.5: An optimiser proposes new parameter values. Gradient descent uses local slope information from the objective. Other solvers can reach the same least-squares solution by different routes.

2.7.2 A Straight-Line Model

Definition 2.7.2.1: A simple linear regression model assigns one predicted value to each input x. The intercept sets the prediction at $x = 0$. The slope states how much the prediction changes when x increases by one unit.

$$\hat{y} = \theta_0 + \theta_1 x \tag{11}$$

Definition 2.7.2.2: The hypothesis is a single pair of parameters that produce predictions for every observation.

$$h\theta(x_i) = \theta_0 + \theta_1 x_1 \tag{12}$$

Definition 2.7.2.3: The residual measures the vertical gap between prediction and observation.

$$e_i = \hat{y}_i - y_i \tag{13}$$

Definition 2.7.2.4: The least-squares cost averages squared residuals across all m training observations.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{j=1}^{m} (h\theta(x_i) - y_i)^2 \tag{14}$$

Definition 2.7.2.5: Each point in parameter space represents one complete line. Moving horizontally changes θ_0 and moving vertically changes θ_1. The height $J(\theta_0, \theta_1)$ records how badly the line fits the data.

2.7.3 Gradient Descent

Definition 2.7.3.1: The gradient vector is defined as follows, where both partial derivatives aare evaluated using the same current parameter values:

$$\nabla J(\theta_0, \theta_1) = \left[\frac{\delta J}{\delta \theta_0}, \frac{\delta J}{\delta \theta_1} \right]^T \tag{15}$$

Adapting the equation, we can get the gradient with respect to the intercept.

$$\frac{\delta J}{\delta \theta_0} = \frac{1}{M} \sum_i h\theta(x_i) - y_i \tag{16}$$

Adapting the equation, we can get the gradient with respect to the slope.

$$\frac{\delta J}{\delta \theta_1} = \frac{1}{M} \sum_i (h\theta(x_i) - y_i) x_i \tag{17}$$

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2.7.4 The Controlled Experiment

When viewing a learning-rate curve we can adapt the hyperparameters based on the emerging patterns.

Pattern	What The Curve Shows	Likely Response
Slow Descent	Cost decreases.	Increase α cautiously.
Smooth Convergence	Cost fails and stabilises.	Retain and verify predictions.
Oscillation	Cost alternates.	Reduce α.
Divergence	Cost grows rapidly.	Reduce α and inspect scaling.

Table 4: Reading the learning-rate curves.

2.8 Linear Regression

2.8.1 Three Routes to the Same Least-Squares Line

The iterative route uses gradient descent. It starts from an initial guess and repeatedly reduces the cost. Its result depends on learning rate, iteration budget, and numerical behaviour.

The direct route uses the closed form to calculate the least-squares parameters from summary quantities. *LinearRegression* packages a robust estimator behind a standard interface.

$$\theta_0 = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sum_i (x_i - \overline{x})^2} \tag{18}$$

2.8.2 Regression Metrics

Mean Square Error:

$$MSE = \frac{1}{m} \sum_i (\hat{y}_i - y_i)^2 \tag{19}$$

Root Mean Squared Error:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{m} \sum_i (\hat{y}_i - y_i)^2} \tag{20}$$

Mean Absolute Error:

$$MAE = \frac{1}{m} \sum_i |\hat{y}_i - y_i| \tag{21}$$

Definition 2.8.2.1: The coefficient of determination compares squared error with the test-set mean baseline.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2} \tag{22}$$

2.8.3 Implementation and Optimisation Variants

Definition 2.8.3.1: Batch gradient descent uses every training row for each update. The path is stable, but each update costs a full pass through the data.

Definition 2.8.3.2: Stochastic gradient descent uses one row per update. Mini-Batch gradient descent uses a small group such as 32 rows. Updates become noisier but more frequent.

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2.9 Multivariate Linear Regression

2.9.1 From One Feature to Many

In multivariate machine learning models, a prediction adds one weighted contribution per feature. With two features, the fitted object is a plane. Higher dimensions follow the same algebra.

Definition 2.9.1.1: The vector hypothesis is defined as:

$$h\theta(\mathbf{x}) = \theta_0 + \theta_1 x_1 + ... + \theta_n x_n = \theta^T \mathbf{x} \tag{23}$$

Definition 2.9.1.2: The design matrix stacks m examples as rows and n features as columns. The resulting matrix $\mathbf{X}$ lets one matrix multiplication produce every prediction at once. We then prepend a bias column where $x_0 = 1$ for every example which lets the intercept participate in the same dot product as the remaining coefficient. Column 0 contains ones, θ_0 multiplies that column, and there is no separate intercept code.

In this matrix, $\mathbf{X}$ is the design matrix including bias $(m \times (n+1))$, θ is the coefficient vector $((n+1) \times 1)$, $\mathbf{y}$ is the observed targets $(m \times 1)$, $\mathbf{X}\theta$ is the predictions $(m \times 1)$, and $\mathbf{X}^T(\mathbf{X}\theta - \mathbf{y})$ is the gradient numerator $((n+1) \times 1)$.

Definition 2.9.1.3: The vectorised cost is the residual vector containing one prediction error per row.

$$J(\theta) = \frac{1}{2m}(\mathbf{X}\theta - \mathbf{y})^T(\mathbf{X}\theta - \mathbf{y}) \tag{24}$$

Definition 2.9.1.4: The vectorised graident is the transpose combining the residual evidence for every coefficient at once.

$$\nabla J(\theta) = \frac{1}{m}\mathbf{X}^T(\mathbf{X}\theta - \mathbf{y}) \tag{25}$$

Definition 2.9.1.5: Multivariate gradient descent initialises θ, computes $\mathbf{X}\theta$, forms residuals, computes $\frac{\mathbf{X}^T e}{m}$, and then updates θ.

2.9.2 Feature Scaling and Optimisation Geometry

When feature scales differ greatly, equal changes in different coefficients produce very different movements in prediction space. Gradient descent can zigzag across a narrow valley, slowing progress.

2.9.3 Direct and Iterative Solvers

The normal equation sets the gradient to zero to produce a linear system for the least-squares coefficients.

$$\theta = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{26}$$

2.10 Polynomial Regression

The prediction curves in x, while every coefficient still enters as a linear multiplier.

$$\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2 + ... + \theta_d x^d \tag{27}$$

In a known synthetic relationship, thirty training observations sample a smooth signal with Gaussian noise. A separate set of 1,000 observations estimates validation error.

$$y = \sin(2\pi x) + \epsilon \tag{28}$$

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2.10.1 Bias

Definition 2.10.1.1: Bias describes systematic error created by restrictive assumptions. Across different training samples, a high-bias model tends to miss the same parts of the true relationship.

Definition 2.10.1.2: Variance describes sensitivity to the particular training sample. A high-variance model changes its fitted curve substantially when the sample changes.

2.11 Regression Loss Functions

2.11.1 What the Model Minimises

Definition 2.11.1.1: A model family states which predictions are possible. The loss states which mistakes matter most. Optimisation then searches for parameters that minimise the average loss.

Definition 2.11.1.2: Positive residuals indicate overpredictions and negative residuals indicate underpredictions, where magnitiude gives size. This can be calculated using:

$$r_i = \hat{y}_i - y_i \tag{29}$$

We can calculate loss using one of the methods specified in Table 5.

Loss	Per-Example Expression	Main Behaviour
Squared	r^2	Quadratic Growth
Absolute	$\|r\|$	Linear Growth
Huber, $\sigma = 1$	$\frac{1}{2}r^2$ near 0, $\|r\| - \frac{1}{2}$ outside.	Smooth and Robust
Log-Cosh	$\log\cosh(r)$	Smooth, nearly linear outside
Quantile, q	$\max(qe, (q-1)e), e = -r$	Asymmetric

Table 5: Methods of calculating loss.

When considering which type of loss to use, it is useful to consider the situation. Some common examples are listed below in Table 6.

Situation	Useful Loss	Reason
Gaussian-like without severe outliers.	Squared	Efficient, smooth, and familiar.
Strong contamination or heavy tails.	Absolute	Limited influence from extreme residuals.
Mostly clean data with some outliers.	Huber/log-cosh	Quadratic near zero and robust in tails.
A conditional percentile is required.	Quantile	Asymmetric loss targets q directly.

Table 6: Common situations and their recommended loss types.

2.11.2 Outliers and Robust Fitting

Definition 2.11.2.1: The squared-loss gradient demonstrates that a large residual produces a proportionally large contribution.

$$\frac{\delta J}{\delta \theta_0} = mean(2r), \frac{\delta J}{\delta \theta_1} = mean(2rx) \tag{30}$$

This produces a smooth update where large errors dominate, and that matches MSE fitting.

Definition 2.11.2.2: The absolute-loss gradient removes the residual magnitude from the derivative, making a bounded contribution that is non-differentiable at zero.

$$\frac{\delta J}{\delta \theta_0} = mean(sign(r)), \frac{\delta J}{\delta \theta_1} = mean(sign(r)x) \tag{31}$$

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Definition 2.11.2.3: The Huber gradient gives the threshold which clips the influence of large residuals.

$$L'\delta(r) = r \text{ if } |r| \leq \delta, \text{ otherwise } \delta sign(r) \tag{32}$$

When fitting a generic robust line, we use the following steps. When doing this, only loss_grad changes between experiments, with representation and gradient-descent loops staying fixed.

1. Initialise θ.
2. Predict $\hat{y}$.
3. Compute r.
4. Evaluate $L'(r)$.
5. Update both θ values.

2.11.3 Quantile Regression

A mean or median gives one central value. Many decisions need a range that changes with x, especially when outcome variability grows across the input space.

Definition 2.11.3.1: The prediction interval describes where a future individual outcome may fall. It includes irreducible observation noise.

Definition 2.11.3.2: The confidence interval for the mean describes uncertainty in the estimated conditional mean. It narrows as the mean becomes more precisely estimated.

When selecting a regression loss model, you should follow these steps:

1. Identify whether extreme residuals represent noise or real risk.
2. Decide whether the target is a mean, median, or another quantile.
3. Inspect the tail growth of $L(r)$.
4. Inspect whether $L'(r)$ is bounded.
5. Check optimisation smoothness and available solvers.
6. Validate behavious with plots, parameters, and held-out evidence.

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

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3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

3.5 Heuristic Functions and Their Development

3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms

3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm

3.8 Applying Different Ant Systems to the TSP

3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

4.1 Introduction to AI Ethics and Related Concepts

4.2 Societal Impacts of AI

4.3 Defining Bias and Fairness in ML

4.4 How to Build Better Systems

4.5 Value Sensitive Design

4.6 Evaluations and Audits

4.7 Bias Mitigation Strategies

4.8 Transparency and Accountability for Trustworthy Systems

4.9 Explainable AI

4.10 Future Trends and Ethical Frontiers

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Part I
Appendix

5 Bibliography

5.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

5.2 References

5.3 List of Figures

5.4 Reading List

5.4.1 Machine Learning

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5.4.2 Ethics and Bias in AI

Various, will be added over time.